GLOBAL SECTIONS ON THE PUNCTURED SPECTRUM VANISH ON A TRUNCATED RANGE: EVERY CLASS IS REPRESENTED BY A DUAL KOSZUL COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-application-nice-K-injective is same-chapter NOT carded (cross-line-claimed label) and is declared external.

Allowed standard prerequisites (editorial).

- Vanishing of quasi-coherent cohomology in high degree on r affines. Cohomology of Schemes, lemma vanishing-nr-affines (cross-chapter, named). H^j(U, F) = 0 for j >= r.
- K-injective truncation comparison. lemma-application-nice-K-injective (same chapter, NOT carded, declared external). H^0(U, E) = H^0(U, tau_{>= -r+1} E).
- Represent classes on the open. proposition-represent-cohomology-class-on-open (carded L26767). Some I_e -> O_X -> E is zero for large e.
- Spectral sequence of a double complex. Derived Categories, lemma two-ss-complex-functor (cross-chapter, named). Computes H^0(U, tau_{>= -r+1}E) from the cohomology sheaves.

Source reference. Original reference 01XI: focused-seeds/geometry/S9-perf2-represent-cohomology-class-on-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 015J: focused-seeds/geometry/S9-perf2-represent-cohomology-class-on-closed/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08D4: sources/stacks-perfect/pinned-original-source.tex, source lines 375–375. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 08DD: sources/stacks-perfect/pinned-original-source.tex, source lines 2316–2316. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 08CZ: sources/stacks-perfect/pinned-original-source.tex, source lines 2116–2116. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 08CZ. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Assume that $H^i(E)|_U = 0$ for $i = -r+1, \ldots, 0$. Then given $s \in H^0(X, E)$ there exists an $e \geq 0$ and a morphism $K_e \to E$ such that s is in the image of $H^0(X, K_e) \to H^0(X, E)$.*

Proof. Since U is covered by r affine opens we have $H^j(U, \mathcal{F}) = 0$ for $j \geq r$ and any quasi-coherent module (Cohomology of Schemes, Lemma 01XI). By Lemma 08D4 we see that $H^0(U, E)$ is equal to $H^0(U, \tau_{\geq -r+1}E)$. There is a spectral sequence

$$H^j(U, H^i(\tau_{\geq -r+1}E)) \Rightarrow H^{i+j}(U, \tau_{\geq -N}E)$$

see Derived Categories, Lemma 015J. Hence $H^0(U, E) = 0$ by our assumed vanishing of cohomology sheaves of E. We conclude that $s|_U = 0$. Think of s as a morphism $\mathcal{O}_X \to E$ in $D(\mathcal{O}_X)$. By Proposition 08DD the composition $I_e \to \mathcal{O}_X \to E$ is zero for some e. By the distinguished triangle $I_e \to \mathcal{O}_X \to K_e \to I_e[1]$ we obtain a morphism $K_e \to E$ such that s is the composition $\mathcal{O}_X \to K_e \to E$. □